SPE-230049-MS

Enhancing ESP Reliability: Synergizing Novel Machine Learning Prediction Model with Operational Practices for Failure Reduction

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper presents a machine learning–driven workflow to enhance reliability in Electrical Submersible Pump (ESP) operations through early risk identification. Deployed across 250 ESPs in a South American asset, the solution integrates real-time sensor data, historical failures, and operational context to detect anomalous behavior and predict failure probability up to 90 days in advance.

The system is composed of two models: a Healthy Model for anomaly detection and a Survival Model to estimate time-to-failure. These models feed into a dashboard that delivers daily alerts and drives timely field actions such as chemical cleanups, parameter adjustments, and surface recalibrations.

The deployment resulted in a 72% recall rate and enabled preventive interventions in 75% of the high-risk wells during Q1 2024. On average, run life was extended by over 145 days in successfully mitigated cases. The paper summarizes key technical learnings and outlines how predictive analytics can support more efficient ESP management.

Introduction

In mature oilfields, Electrical Submersible Pumps (ESPs) are the primary artificial lift solution, representing over 94% of completions in the asset under study. Their uninterrupted performance is vital to maintain production and manage operational expenditures. Field teams monitor ESP behavior using real-time data transmitted every 20 minutes, allowing for immediate responses to abnormal events.

However, many failure modes evolve gradually and are difficult to detect using reactive diagnostics or threshold-based alarms alone. Subtle degradation patterns can remain unnoticed until production losses or equipment shutdowns occur. This creates a need for solutions that complement existing monitoring workflows with earlier and deeper visibility into equipment health.

To address this, a machine learning–based workflow—referred to as the ESP Health Checker—was developed and field-tested. It combines engineered features from historical and live operational data to detect behavioral deviations and estimate failure likelihood. Rather than replacing current surveillance, it augments it by generating actionable insights that support preventive actions such as surface tuning, chemical treatments, or equipment calibration. This paper presents the design, deployment, and field results of the solution, demonstrating how it helps bridge the gap between monitoring and proactive reliability management.

Recent Technology progress for surveillance

Conventional Practices

The monitoring of Electrical Submersible Pumps (ESPs) has undergone a significant evolution. Traditionally, engineers relied on offline inspection of amperometric charts from variable speed drives to assess ESP health. These threshold-based methods, while helpful, typically trigger alarms only when operating limits are exceeded—often when failure is already imminent (Lastra and Xiao 2022). With the introduction of real-time telemetry platforms, ESP behavior can now be monitored continuously, allowing operators to view surface and downhole parameters such as motor current, discharge pressure, and fluid rate every few minutes. However, this increased data availability still demands manual interpretation and expert review, which may not capture subtle degradation trends early enough to prevent failure (Devshali et al. 2022). In response, the industry has increasingly turned to machine learning (ML) solutions that can learn from historical data and recognize abnormal patterns proactively.

Recent Advances

In recent years, machine learning (ML) has gained traction as a valuable tool for improving Electrical Submersible Pump (ESP) reliability. Unlike traditional monitoring—which typically waits for a single variable like motor current to exceed a fixed limit before triggering an alarm—ML models learn from the combined behavior of many parameters and detect unusual patterns before a failure occurs. This allows field engineers to act earlier, extending equipment life and minimizing unplanned shutdowns.

For example, instead of monitoring only motor current, an ML model might analyze how current, vibration, intake pressure, and fluid rate interact over time. If these signals start to behave differently from what's been seen during healthy operation, the model raises an alert—much like an experienced engineer would notice "something isn't quite right" even if no alarm has sounded. This concept is illustrated in Figure 1, which compares traditional threshold logic to ML-based anomaly detection.

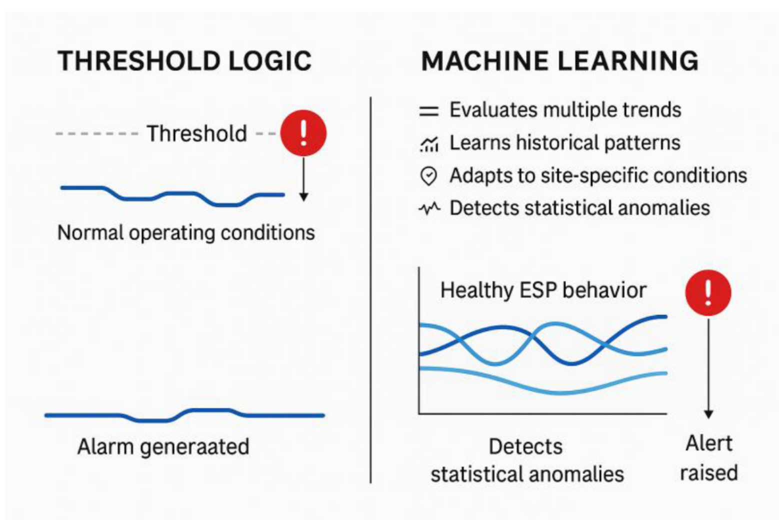


Figure 1—Comparison of threshold-based monitoring and machine learning for ESP surveillance. While threshold logic triggers alarms only after fixed limits are breached, ML evaluates multiple trends and recognizes early degradation patterns for predictive action.

Supervised ML models like Random Forest and Gradient Boosting are among the most used. These models require historical examples of past failures to learn from and are especially effective when companies have labeled datasets that describe what a failure looks like in terms of sensor behavior (Aslam et al. 2023; Devshali et al. 2022). On the other hand, when failure records are limited, unsupervised models—such as Isolation Forests or Autoencoders—can learn what normal performance looks like and then identify deviations without needing failure labels (El Sherbeny et al. 2024).

Beyond early warnings, some workflows aim to estimate how much life is left in the ESP. This is known as Remaining Useful Life (RUL) prediction. Some authors apply Weibull distributions, a method familiar from reliability engineering, while others use LSTM networks, a type of deep learning suited to time-series data (Hernandez et al. 2024; Lastra, Worth, and Swaffield 2023). These approaches help forecast whether a pump is likely to fail within the next 30, 60, or 90 days.

In advanced cases, ML predictions are embedded in digital twins—virtual replicas of ESP systems that combine physics-based simulations with real-time data. This helps engineers not only see what might go wrong, but also understand why and how to respond (Lastra, Worth, and Swaffield 2023; Hernandez et al. 2024). Several field applications now feature interactive dashboards where engineers can view failure probabilities, contributing factors, and suggested actions—all in real time (El Sherbeny et al. 2024; Hernandez et al. 2024). These tools support a shift from reactive alarms to predictive, decision-support systems tailored to real operating conditions.

Gaps in Implementation

While recent advances in machine learning (ML) have shown strong potential for enhancing ESP reliability, several gaps remain between algorithm development and real-world application. One of the main challenges is the **limited field validation** of many proposed solutions. Although numerous studies demonstrate high performance on historical or synthetic datasets, fewer models have been tested in live operations where conditions are dynamic and sensor data can be noisy or incomplete (Khabibullin et al. 2020; Lastra, Worth, and Swaffield 2023). Bridging this gap requires robust deployment of pipelines, ongoing retraining, and consistent model monitoring.

A second barrier is the **operational integration** of ML outputs. In some cases, predictive alerts remain isolated from day-to-day engineering decisions due to poor interpretability or misalignment with existing workflows—such as maintenance planning, rig scheduling, or chemical treatment campaigns (Aslam et al. 2023; Karnik et al. 2021). If the information provided by the model is not clear, timely, or actionable, it risks being ignored or deprioritized by field teams who operate under tight constraints.

Finally, **trust and adoption** continue to be critical hurdles. Many field engineers express skepticism toward "black box" models that provide predictions without transparent logic or a clear physical basis. To gain acceptance, ML systems must evolve toward **explainable outputs**, where alerts are accompanied by context (e.g., sensor combinations, deviation magnitude, historical match) and recommendations are framed in operational terms. Recent field deployments have shown that when alerts are delivered through familiar platforms and aligned with engineer workflows, the models are more readily adopted and deliver real impact (El Sherbeny et al. 2024; Hernandez et al. 2024).

Methodology

This section outlines the methodology used to develop, validate, and operationalize a machine learning (ML) solution for identifying **anomalous ESP behavior** and estimating the probability of failure. The workflow includes data gathering, QA/QC validation, feature engineering, model training, and integration into daily production operations.

Data Sources

The foundation of this project was a multi-source dataset compiled over four years (January 2020 to July 2024) from a mature South American oilfield, where over **654 ESP installations** were deployed across **266 active wells**. The data originated from four key sources: (1) high-frequency ESP sensor telemetry, (2) failure and downtime logbooks, (3) well-level production data, and (4) laboratory measurements. These inputs, generated at varying periodicities, were consolidated and aligned to form a unified timeline for model training and evaluation.

1. ***ESP Telemetry***

   Sensor data was collected from both surface and downhole equipment at frequencies between 10 and 20 seconds. The original master dataset contained over 30 telemetry variables, including:

   - **Pressures**: Intake and discharge pressure
   - **Temperatures**: Intake temperature and motor temperature
   - **Electrical metrics**: Average amperage, drive frequency, active/passive current leakage, input/output voltage, VSD output current, zero current, and motor load
   - **Vibration:** [GK1][EB2]Measured by downhole accelerometers to monitor oscillations in pump and motor components as an indicator of mechanical condition.
   - **Data quality flags**: Each sensor measurement was accompanied by a binary quality indicator ("GOOD"/"BAD")

   These high-frequency time-series were aggregated to 20-minute intervals during preprocessing to enable scalable processing and minimize noise.

2. ***Failure and Downtime Records***

   Two independent logbooks were combined to generate labeled failure and downtime datasets. The **Failures Dataset** included:

   - Well and ESP installation identifiers
   - Start date (running date) and failure (stop) date
   - Pulling date and total run life (days)
   - Failure classification at component and subcomponent level
   - Root cause and mechanism descriptors
   - Event type (e.g., failure, manual shutdown, active) and contextual comments

   The **Downtimes Dataset** recorded daily events and production losses for each well, specifying event duration, cause classification across three levels, and whether root cause identification was achieved.

3. ***Production and Lab Data***

   Monthly and weekly **production data** was merged from three different sources: manual well tests, surface metering, and internal field monitoring systems. Variables included:

   - **Volumes**: Total fluid (BFPD), oil (BOPD), water (BWPD)
   - **Properties**: Water cut (BSW), gas-oil ratio (GOR), salinity, API gravity
   - **Frequency**: Reported pump frequency during test

4. ***Laboratory measurements***

- Emulsion percentage, free water, and solids content (ppm)
- Solid type, salinity, water hardness, and emulsion
- Sample comments and test method metadata

These datasets were temporally matched to the ESP signal history using timestamp alignment and reverse-filling techniques. Together, they enabled the development of robust predictive models by linking sensor patterns with operational and production events under real conditions.

Data Preparation

After collecting data from sensors, failure records, downtimes, and production tests, the next step was to clean, align, and organize all information into a single dataset. This preparation was key to ensuring the model learned from reliable, real-world field conditions.

1. ***Standardization and Synchronization***

   Each data source had different formats and frequencies. Sensor data came every few seconds, while production and failure records were monthly. To make these compatible:

   - All sensor data was **resampled to 20-minute intervals**, providing a regular and manageable time base.
   - **Invalid values and gaps**—such as missing sensor readings or out-of-range measurements—were filtered or filled using smart techniques like reverse filling and interpolation.
   - Production records were **standardized** to match well names and timestamps from the sensor data.

   This allowed us to build a clear timeline for each pump's behavior.
2. ***Event Alignment***

   To properly train the models, it was important to know when a pump was operating normally, shut down, or had failed. For this:

   - **Failure and downtime logs** were processed to mark exactly when each event started and ended.
   - These time tags were added to the sensor data, creating a full operational history for each pump.
   - During failures or planned shutdowns, interpolated values (such as estimated production) were removed to avoid misleading the model.
3. ***Final Dataset***

   The result was a single, labeled dataset where every data point includes:

   - Sensor readings (pressures, temperatures, electrical load, vibration)
   - Production values (oil, water, total fluid, GOR, API)
   - Operational status (normal, downtime, or failure)

This unified dataset formed the basis for building models that understand what "healthy" operation looks like—and detect when a pump starts to behave differently.

Feature Engineering

To enhance the model's ability to detect early signs of abnormal pump behavior, a comprehensive feature engineering process was applied to the cleaned dataset. This step transformed raw sensor values and operational data into more meaningful variables that could highlight patterns not immediately visible in individual readings.

The feature engineering strategy focused on three main areas: **domain-specific transformations**, **statistical summaries**, and **time-lagged behavior**.

1. ***Operational and Contextual Features***

   Several new variables were created to give the model more insight into the physical behavior and lifecycle of each ESP installation:

   - **Delta pressure** was calculated as the difference between discharge and intake pressures, serving as a proxy for the pump's hydraulic performance. Values below 1,000 psi were excluded to avoid noise.
   - **Runlife and downtime metrics**, including total runtime in days, number of prior downtimes, total accumulated downtime, and mean time between shutdowns, were computed to capture operational stress.
   - **Time since telemetry loss** was tracked as an additional indicator of equipment stability or communication gaps.

   These features gave the model a broader understanding of how long and under what conditions each ESP had been operating.
2. ***Statistical Trend Features***

   Sensor behavior over time was analyzed using **rolling window statistics**, computed per well:

   - Rolling **mean, maximum, minimum, and standard deviation** were calculated for key variables such as motor current, temperatures, pressures, and vibration.
   - Two time windows were applied: short-term (e.g., 1 hour) and long-term (e.g., 24 hours).
   - These helped the model identify gradual changes—such as a slow rise in motor temperature or drop in intake pressure—that may indicate deterioration.
3. ***Time-Lag Features***

   To help the model recognize changes compared to recent history, **lag features** were created:

   - Each key sensor signal was shifted backward in time to capture its value at previous steps.
   - These lagged values served as a reference for how a parameter evolved, mimicking how engineers review trendlines day-to-day.
4. ***Aggregation for Failure Risk Estimation***

   For the survival model—which estimates the probability of failure within a future time window—all 20-minute data was **aggregated to monthly summaries**. These included both raw and engineered features, allowing the model to evaluate long-term degradation patterns without overfitting to noise.
5. ***Feature Selection***

   To reduce complexity and improve explainability, the final set of features was selected based on:

   - **Statistical correlation**, removing highly redundant inputs
   - **Engineering expertise**, prioritizing variables known to influence ESP performance

Machine Learning Models

The modeling workflow was structured around two complementary goals:

1. **Anomaly Detection** – to simulate healthy pump behavior and flag deviations in real time.

2. **Survival Model – Failure Probability Estimation** – to forecast the probability of failure within a future time window using survival analysis.

These models were trained using the engineered features described in the previous section, which captured both sensor behavior and operational history.

Anomaly Detection – Healthy Model

To help detect early signs of failure, we first created models that simulate how an ESP behaves when it is working properly. These models were designed to predict three key sensor signals:

- **Average amperage**
- **Motor temperature**
- **Delta pressure** (difference between discharge and intake pressure)

Lasso Regression (Least Absolute Shrinkage and Selection Operator) is a linear modeling technique that includes an L1 regularization term, which penalizes the absolute size of the regression coefficients (Tibshirani, 1996). This has the effect of shrinking the coefficients of less important variables to zero, automatically performing feature selection. The result is a simpler and more interpretable model that avoids overfitting in high-dimensional time series data.

For each sensor, we trained a model that learns from the sensor's recent behavior—specifically, how it has performed over the last 8 hours. This includes four key indicators:

- The **average** value
- The **lowest** value
- The **highest** value
- The **variation** (standard deviation)

These indicators were calculated for each signal using its past values, helping the model understand both the level and the variability of the pump's operation over time.

Once the models were trained, they were used to predict what the sensor values *should* look like if the ESP was operating under normal, healthy conditions. These predictions serve as a reference or "digital twin" for comparison.

Then, for each data point, we compared the **actual value** from the field with the **predicted healthy value**. The difference between them is what we call the **anomaly**:

Anomaly = Actual Value – Predicted Healthy Value

If this difference is close to zero, the pump is likely to operate normally. If the difference is large, it may indicate that the pump is starting to deviate from healthy behavior.

In addition to these differences, we also calculated **cumulative anomalies**—which add up the small deviations over time. This helps highlight pumps that may be gradually deteriorate, even if individual differences are small.

The result was a new dataset with six indicators:

- Three direct anomalies (for amperage, temperature, and pressure)
- Three cumulative anomalies.

This enriched dataset was then used to support the second part of the analysis: estimating the probability of failure, which is explained in the next section.

Model accuracy was evaluated using the following metrics:

- **R^2 (Coefficient of Determination)** quantifies the proportion of variance in the observed values that is predictable from the input features. An R^2 value of 1.0 indicates perfect prediction, while 0 suggests no predictive power. This metric is widely used in regression tasks to assess how well a model fits the data (Kutner et al., 2004).
- **MAPE (Mean Absolute Percentage Error)** provides a scale-independent measure of error, calculated as the mean of the absolute percentage differences between predicted and actual values. It is especially helpful when comparing model accuracy across sensors with different scales or units (Armstrong, 1985).
- **Pearson Correlation Coefficient (r)** measures the strength and direction of the linear relationship between the predicted and actual values. A value of 1 indicates perfect positive correlation, -1 indicates perfect negative correlation, and 0 implies no linear correlation. This metric is often used to validate the consistency of time-series predictions (Benesty et al., 2009).

Survival Model – Failure Probability Estimation

To estimate the risk of ESP failure over time, a **Random Survival Forest (RSF)** model was used. RSF is an ensemble learning method designed for **time-to-event problems**, particularly well-suited for field data where some pumps are still running and haven't failed—known as **right-censored data**.

A **(RSF)** algorithm was selected due to its ability to:

- Handle nonlinear relationships between input variables,
- Manage high-dimensional datasets with a mixture of numeric and categorical features,
- Incorporate censored observations without biasing the failure risk estimation.

Each tree in the RSF ensemble produces an estimate of the **survival function**, S(t)S(t)S(t), which represents the probability that a given ESP remains operational beyond time. These estimates are aggregated across the forest to provide a robust prediction for each pump instance.

Data Preparation

The input dataset was constructed by merging:

- Engineered features from ESP sensor readings (e.g., motor temperature, pressure, vibration),
- Anomaly features from the Healthy Model (i.e., deviation from expected behavior),
- Operational metadata (downtime history, run life, accumulated failure indicators).

To structure the data for monthly prediction intervals, features were aggregated over calendar months, enabling the model to learn risk patterns on a temporal basis. Pumps with insufficient operational records or inconsistent sensor data were excluded from the training dataset.

Key steps in the data transformation pipeline included:

- Calculation of **time-to-failure** for each pump instance,
- Labeling of failure vs. censored (no-failure) observations,
- Engineering of metadata such as **mean time between downtimes (MTBD)** and **accumulated anomaly durations**,
- Computation of survival targets (event flag and time duration until failure or censoring).

Model Training and Validation

The dataset was divided into training, validation, and test sets using **stratified time-aware partitioning**, ensuring temporal and field-wise consistency across the subsets. Pumps used for testing were fully excluded from model training to simulate real-world deployment on unseen wells.

During training, the RSF learned to associate patterns in the engineered features with the likelihood of pump survival or failure. The model's ability to **rank wells by failure risk** was a key objective, enabling operators to prioritize maintenance resources.

Deployment Considerations

The survival model was integrated into a digital pipeline that allows for:

- Continuous ingestion of new sensor data,
- Periodic model retraining as new failures are observed,
- Visualization of predicted failure probability across all active wells.

This implementation supports **early warning of high-risk pumps**, improves planning for rig and pull schedules, and enhances field reliability by enabling a shift from reactive to preventive ESP management.

Performance was assessed using:

- **Concordance Index (C-index)**: Measures the model's ability to correctly rank pumps by risk. A score of 1.0 indicates perfect ranking, while 0.5 is no better than random.
- **Precision, Recall, and F1-score**: Calculated from binary risk scores (e.g., "high failure probability within 90 days"), these metrics quantify how well the model detects true positives without excessive false alarms.

Both models were developed and deployed in an integrated environment, enabling **continuous monitoring, model versioning, and retraining** as new data becomes available. Together, the Healthy Model and the Survival Model form the core of the ESP Health Checker system—supporting proactive decisions, optimizing resource allocation, and reducing unplanned ESP downtime.

Results and Analysis

Anomaly Detection – Healthy Model

To simulate the expected behavior of (ESPs) under stable operating conditions, three independent Lasso Regression models were developed. These models were trained to predict key telemetry signals—average amperage, motor temperature, and delta pressure—based solely on their own lagged historical behavior. By comparing predicted values against real-time sensor measurements, residuals were computed to serve as indicators of anomalous behavior.

The predictive performance of these models was assessed using the coefficient of determination (R^2), Mean Absolute Percentage Error (MAPE), and Pearson correlation coefficient. The **Average Amperage model** demonstrated excellent accuracy, with an R^2 of **0.9988**, MAPE of **0.50%**, and Pearson r of **0.9994**, indicating strong linear agreement between predicted and observed data. Similarly, the **Motor Temperature model** achieved an R^2 of **0.9941**, MAPE of **0.21%**, and Pearson r of **0.9971**. The **Delta Pressure model** reported R^2 = **0.9917**, MAPE = **0.38%**, and Pearson r = **0.9958**.

These results confirm the models' capability to accurately replicate healthy behavior for each monitored parameter. The anomaly residuals and their cumulative versions, derived from model predictions, were subsequently used as inputs for the survival analysis, offering interpretable early-warning signals for potential ESP degradation.

Survival Model – Failure Probability Estimation

To estimate the time-dependent risk of ESP failure, a Random Survival Forest (RSF) model was implemented. This model was trained using a dataset composed of engineered features from ESP sensors, operational metadata, and the anomaly scores generated by Healthy Models. The survival model accounts for censored data, i.e., pumps that had not yet failed, and learns from both failure and non-failure histories to estimate failure probability over a predefined prediction horizon.

Three RSF models were trained using different configurations of input features and pump partitions. All models were evaluated over a **60-day prediction horizon**, with thresholds for failure probability calibrated to optimize the trade-off between sensitivity and precision. The main evaluation metrics include precision, recall, F1-score, and overall accuracy.

Model 1, evaluated at a 30% failure probability threshold, achieved a **precision of 0.88**, indicating that nearly 9 out of 10 pumps flagged as at risk did indeed fail. The **recall was 0.71**, showing that 7 out of 10 failures were correctly identified. The model also reached an **F1-score of 0.79** and **overall accuracy of 0.99**.

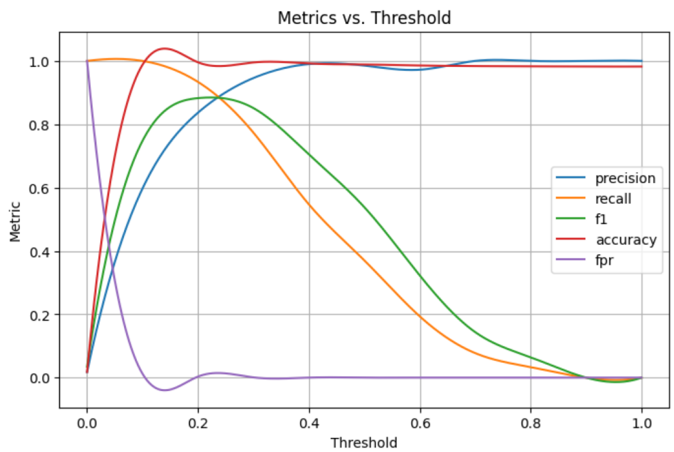


Figure 2—Model 1 – Performance Metrics vs. Threshold (Optimal Threshold = 30%, Prediction Horizon = 60 Days)

Model 2 showed a slightly lower precision of **0.76**, but maintained a similar recall at **0.73**, with an F1-score of **0.74**, and accuracy remaining at **0.99**. Its optimal classification threshold was identified at 20% failure probability.

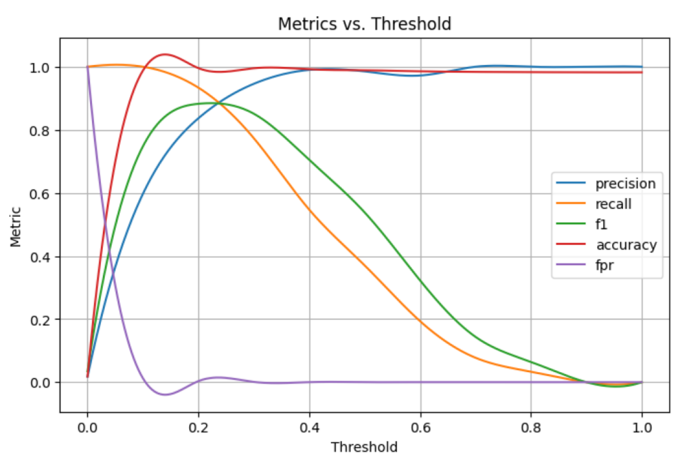


Figure 3—Model 2 – Performance Metrics vs. Threshold (Optimal Threshold = 30%, Prediction Horizon = 60 Days)

Model 3, trained on an alternative subset of pumps, yielded results consistent with Model 1: **precision of 0.88**, **recall of 0.71**, **F1-score of 0.79**, and **accuracy of 0.99**, using the same 30% probability threshold.

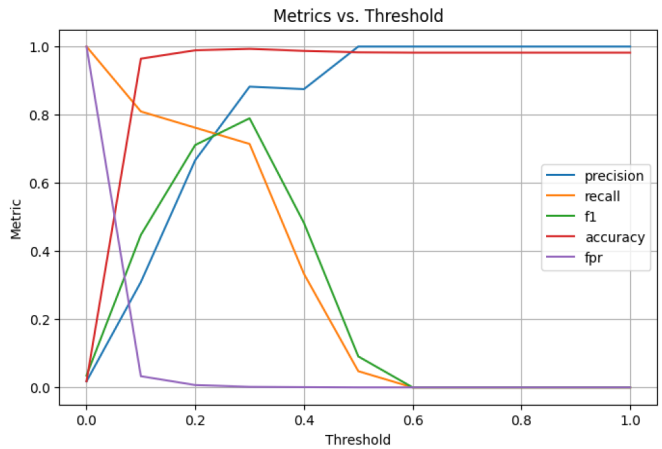


Figure 4—Model 3 – Performance Metrics vs. Threshold (Optimal Threshold = 30%, Prediction Horizon = 60 Days)

Overall, the survival models demonstrated robust classification performance, with high accuracy and consistent precision-recall balance. The results suggest that the system can effectively distinguish pumps with elevated failure risk, supporting data-driven prioritization of preventive maintenance and resource allocation across operational assets.

Operational Integration

Prior to the implementation of this tool, ESP monitoring relied solely on traditional surveillance methods. After well completion with ESPs, engineers monitored downhole and surface parameters to detect abnormalities. When deviations were noticed—such as pressure fluctuations, temperature rises, or current instability—corrective actions were recommended, including equipment diagnostics or chemical treatment adjustments. While effective to a degree, this process was reactive, and early signs of failure were sometimes missed.

The new machine learning-based monitoring workflow introduced a structured process to support early detection of failure risk and trigger preventive actions, as illustrated in Figure 5. The workflow includes the following steps:

1. **High Failure Probability Detection**: The system generates a daily risk ranking of active ESPs. Pumps predicted to have a high probability of failure are prioritized for engineering review. This ranking is visualized through an interactive dashboard (see Figure 6) that displays the predicted failure probability alongside production indicators (e.g., BOPD, runlife).
2. **Trend Analysis and Engineering Review**: Surveillance engineers perform a deeper analysis of high-risk wells using conventional diagnostic tools and monitoring plots. Based on findings, they decide whether additional measurements—such as electrical diagnostics or downhole sensor checks—are required.
3. **Remedial Plan Execution**: Possible mitigation actions include VSD optimization, ESP cleanup jobs (tubing or casing), surface equipment adjustments, or well interventions. These decisions are based on a combined evaluation of ML predictions and traditional surveillance.
4. **Post-Action Verification**: After actions are taken, the evolution of the failure probability is continuously tracked by the system. Verification of reduced risk is confirmed using both the ML model and traditional surveillance plots.
5. **Closed-Loop Learning**: Weekly collaboration sessions between engineers and data scientists support model refinement and field adoption. Confidence in the predictions increased as flagged wells consistently exhibited signs of degradation or failure.

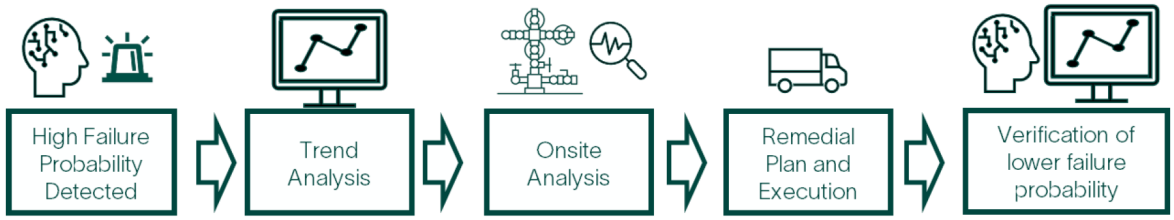
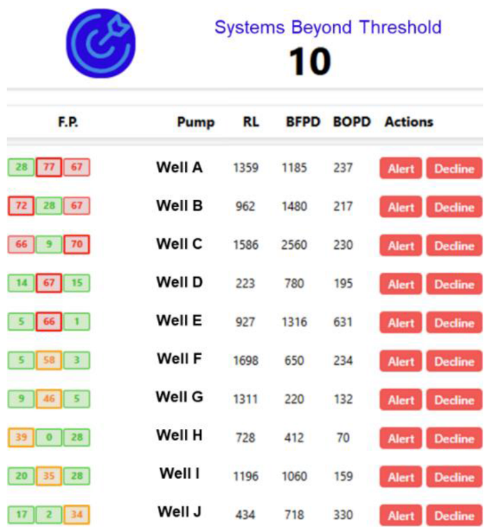


Figure 5—Well Production Monitoring Workflow Using the Machine Learning Workflow



F.P.	Pump	RL	BFPD	BOPD	Actions
28 77 67	Well A	1359	1185	237	Alert Decline
72 28 67	Well B	962	1480	217	Alert Decline
66 9 70	Well C	1586	2560	230	Alert Decline
14 67 15	Well D	223	780	195	Alert Decline
5 66 1	Well E	927	1316	631	Alert Decline
5 58 3	Well F	1698	650	234	Alert Decline
9 46 5	Well G	1311	220	132	Alert Decline
39 0 28	Well H	728	412	70	Alert Decline
20 35 28	Well I	1196	1060	159	Alert Decline
17 2 34	Well J	434	718	330	Alert Decline

Figure 6—Failure Probability Ranking Dashboard for Daily Prioritization

Case Study – Operational Application of the ESP Failure Prediction System

To support timely decision-making, a structured operational workflow was established for the weekly review of high-risk ESPs identified by the failure prediction tool. Every Friday or Sunday, an automated email is sent to field engineers, listing the most critical wells based on their failure probability ranking. The emails include visual dashboards and supporting indicators that facilitate early action (see Figure 6.)

When a well is flagged, the engineering team initiates a standardized five-step review process:

1. **Review model output**: Evaluate the predicted failure probability and its historical trend.
2. **Review well intervention history**: Check past rigless operations using internal databases.
3. **Evaluate chemistry performance**: Analyze historical emulsion and solids issues.
4. **Verify ESP parameters**: Examine electrical and hydraulic signals (e.g., amperage, delta pressure).
5. **Define and execute actions**: Coordinate field response according to the technical findings.

Example: Well-163

As shown in Figure 7, Well-163 was flagged on January 26, 2025, with a failure probability of 35%, based on the prediction from Model 2. The tool had previously identified a steep increase in risk, reaching a peak above 50% around day 270 of runlife. A subsequent decline in probability followed, likely linked to stabilizing operating conditions or minor adjustments. However, nearing day 350, the model signaled a renewed rise in failure probability, reaching 35% by the time the alert was issued.

Following the alert:

- A full review of recent ESP performance parameters was initiated, confirming increasing intake pressure and changes in load profiles.
- Chemical injection and solids control data were reviewed to validate flow assurance integrity.

- A review of prior interventions was performed, confirming no recent mechanical adjustments.
- Based on these inputs, a corrective plan was implemented, including degassing optimization and a chemical flush.

Figure 7—Predicted Failure Probability Trend for Well-163 Over ESP Run life

Post-intervention monitoring will determine if the corrective measures result in a reduction in failure probability and extend the ESP runlife.

This case demonstrates how predictive analytics can complement traditional surveillance by flagging subtle, cumulative changes in pump behavior. It enabled early engineering intervention before an actual failure occurred, reducing unplanned downtime risk.

Case Study – Six-Month Summary of Field-Wide ESP Failure Prevention

In a six-month deployment period, the ESP failure prediction tool was integrated into daily operations across 246 active wells. Of these, 66 wells (27%) were flagged by the system for elevated risk, prompting targeted engineering responses. Among the wells flagged, six were identified with run life (RL) below 180 days, typically associated with infant mortality failures.

Following early alerts, more than 60 mitigation actions were executed, including acid stimulation, annular solvent injections, and chemical adjustments. These proactive interventions successfully prevented failures in all six high-risk wells, resulting in an average runlife extension of 150 days.

Beyond the infant failure category, the model also guided actions for 16 intermediate-aged wells (180 < RL < 900 days) and 44 wells with extended runlife (>900 days), achieving successful intervention rates of 87% and 93%, respectively.

Overall, the predictive workflow not only enhanced decision-making and intervention planning but also avoided an estimated 12,000 barrels of deferred oil production, validating the operational and economic impact of the system.

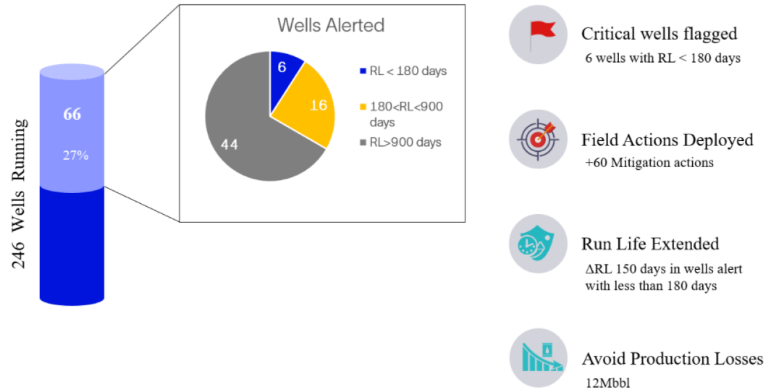


Figure 8—Operational Impact of ML-Based ESP Failure Prediction: 6-Month Field Summary

Conclusion and Future Work

This work demonstrates the successful development and deployment of a machine learning system for predicting Electric Submersible Pump (ESP) failures in a mature field. The two-tiered approach—based on a Healthy Model for anomaly detection and a Survival Model for failure probability estimation—proved effective in capturing early signals of equipment degradation.

Across the models evaluated, precision values reached up to **88%**, with **F1-scores of 0.79** and **overall accuracies of 99%**, confirming the system's ability to detect high-risk pumps with minimal false alarms. These predictive insights supported timely engineering interventions, avoiding failures in **100% of the wells flagged with runlife below 180 days**, and extending their operational life by an average of **150 days**.

Beyond its predictive accuracy, the solution demonstrated tangible operational value:

- **Proactive risk mitigation** reduced downtime and preserved production.
- **Targeted field actions** were executed more efficiently using ranked risk scores and diagnostic dashboards.
- **Trust in data-driven workflows** grew among engineers as model predictions aligned with field performance.
- Sufficient lead time allocated for ESP inventory planning

Looking ahead, future improvements will focus on:

- **Real-time data ingestion and automatic alerting**, enabling continuous monitoring and faster response.
- **Expansion to other lift systems**, such as rod pumps, and deployment across additional geographic regions.
- **Enhanced model explainability and feedback integration**, to further drive adoption and support decision-making in the field.

By integrating predictive models into routine surveillance workflows, this approach represents a scalable path toward more reliable and efficient ESP operations in complex field environments.

Acknowledgements

We acknowledge the SLB management for their permission and support of this work and the publication permission. We acknowledge the contribution our field engineers, data scientists, and platform developers who made this deployment possible.

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